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4.4.8 The relationship between Average Cost and Marginal cost

There is a unique relationship between the AC and MC curves as shown in diagram 4.8.

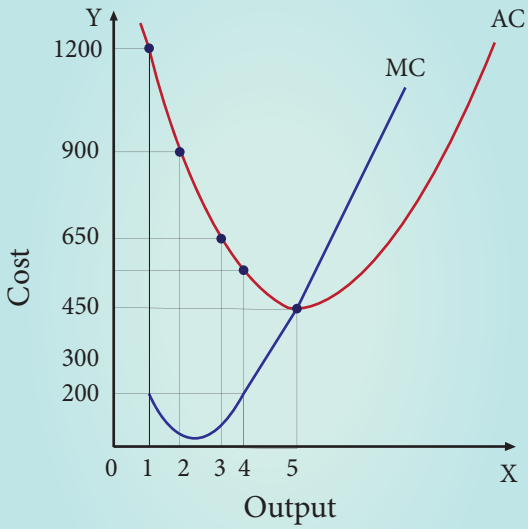


Diagram 4.8

1. When AC is falling, MC lies below AC.
2. When AC becomes constant, MC also becomes equal to it.
3. When AC starts increasing, MC lies above the AC.
4. MC curve always cuts AC at its minimum point from below.

4.5 Long Run Cost Curve:

In the long run all factors of production become variable. The existing size of the firm can be increased in the case of long run. There are neither fixed inputs nor fixed costs in the long run.

LAC is given in diagram 4.9.

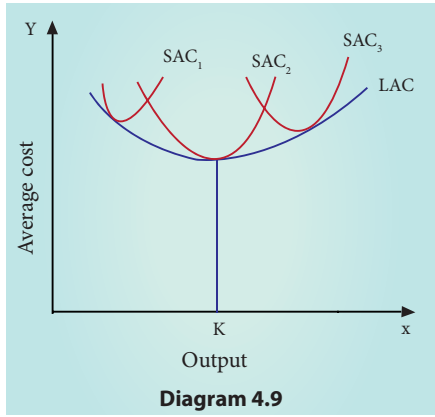


Diagram 4.9

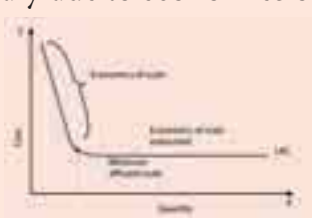
Long run average cost (LAC) is equal to long run total costs divided by the level of output.

$$LAC = LTC/Q$$

where, LAC denotes Long-Run Average Cost,

LTC denotes Long-run Total Cost and

Q denotes the quantity of output.

The LAC curve is derived from short-run average cost curves. It is the locus of points denoting the least cost curve of producing the corresponding output. The LAC curve is called as 'Plant Curve' or 'Boat shape Curve' or 'Planning Curve' or 'Envelop Curve'.

A significant recent development in cost theory is that the long-run average cost curve is L- shaped rather than U-shaped. The L-shape of the long-run average cost curve implies that in the beginning when output is expanded through increase in plant size and associated variable factors, cost per unit falls rapidly due to economies of scale.